USING PREDICTED PROBABILITIES AS COVARIATES IN CAUSAL MEDIATION ANALYSIS

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CAUSAL MEDIATION: THE PROBLEM

- Causal mediation analysis is an approach that attempts to estimate the pathways from an upstream variable to a downstream variable.
- Usually, it is seeking to determine what variable (mediator) a treatment or exposure (upstream variable) is affecting that in turn affects the endpoint (downstream variable).
  - Treatment → mediator → endpoint
- The percent change in the treatment parameter estimates when the mediator is controlled for is an estimate of the percent mediation.
- The procedure assumes that all confounders are controlled for, which is either unrealistic or makes for a very large model in observational studies.
- Additionally, usually only one mediator can be assessed in the analysis.

CAUSAL MEDIATION: PROC CAUSALMED

- SAS has a procedure, PROC CAUSALMED, that fits all the models and estimates the effects (the pathways and percent mediation) along with p-values and Wald confidence intervals.
- However, it only allows binary or continuous treatment and mediator variables, though covariates can be continuous, binary, or multi-category.
- There are other more computationally intensive alternatives, such as PROC CALIS, but these are impractical for very large data sets as they take too long to run or converge.
- The technique presented today is a way of estimating the total mediation from several mediators at once.

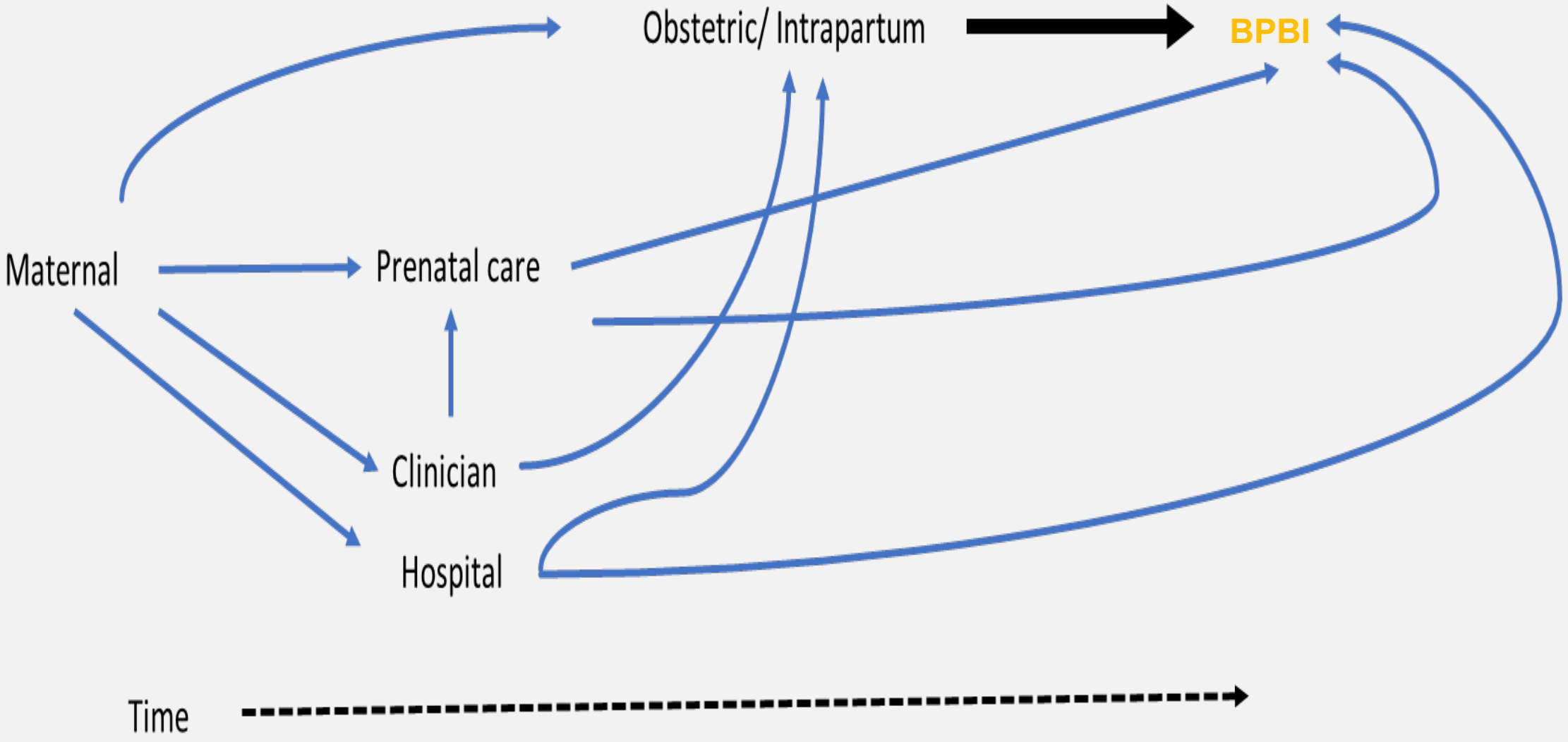
PREDICTED PROBABILITIES FROM LOGISTIC REGRESSION ANALYSIS

- The procedure is:
  - Fit a logistic regression model for the endpoint of interest using all potential mediators of interest, except the exposure or most upstream variable.
  - Save the predicted probabilities for each subject.
  - Run the causal mediation analysis on the exposure and endpoint variables of interest, using the predicted probabilities as the mediator variable.
  - The estimate of the percent mediated is then an estimate of the joint mediation by all covariates.

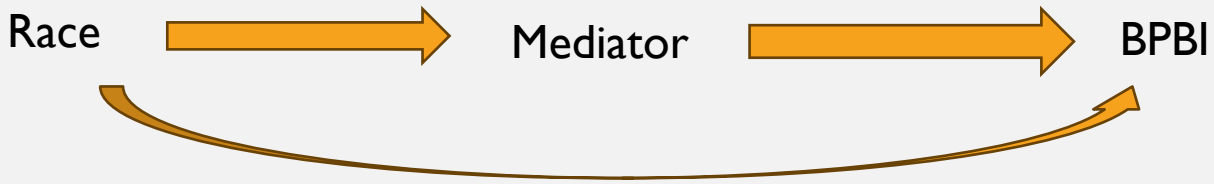
- Brachial plexus birth injury (BPBI)
  - Nerve injury → upper extremity paralysis, musculoskeletal deformity
  - Long term consequences into adulthood
  - Incurable condition despite surgery and rehabilitation
  - Very low incidence, about 1.2 per thousand births, requiring large data sets for estimation.
  - Conventional wisdom posits a pathway of Diabetes/obesity → macrosomia → shoulder dystocia → BPBI.
    - But only about 25% of BPBIs have these upstream factors and 50% have no known risk factors.
- **Imperative to understand BPBI risk factors, especially those that could lead to prevention**



AN EXAMPLE: BRACHIAL PLEXUS BIRTH INJURY


Obstetric/ Intrapartum
BPBI
Maternal
Prenatal care
Clinician
Hospital
Time

Indirect Effect



Direct Effect

```
title "Mother's Race and DM Mediator, controlling for covariates";
PROC CAUSALMED DATA=live.live95 pALL ALPHA=.05;
CLASS live_bpi (descending) mhisp (descending) black (descending) dm
(descending) mage_imp3 (descending)
parity2 (ref='0');
MODEL live_bpi = black dm;
MEDIATOR dm  = black;
covar mage_imp3 mhisp parity2;
run;
quit;
```

SAS OUTPUT FOR DIABETES MEDIATOR

Summary of Effects						
	Estimate	Standard Error	Wald 95% Confidence Limits		Z	Pr > |Z|
Odds Ratio Total Effect	1.6088	0.07311	1.4655	1.7521	8.33	<.0001
Odds Ratio Controlled Direct Effect (CDE)	1.5950	0.07248	1.4530	1.7371	8.21	<.0001
Odds Ratio Natural Direct Effect (NDE)	1.5950	0.07248	1.4530	1.7371	8.21	<.0001
Odds Ratio Natural Indirect Effect (NIE)	1.0086	0.000974	1.0067	1.0106	8.88	<.0001
Total Excess Relative Risk	0.6088	0.07311	0.4655	0.7521	8.33	<.0001
Excess Relative Risk Due to CDE	0.5856	0.07136	0.4458	0.7255	8.21	<.0001
Excess Relative Risk Due to NDE	0.5950	0.07248	0.4530	0.7371	8.21	<.0001
Excess Relative Risk Due to NIE	0.01379	0.001667	0.01052	0.01706	8.27	<.0001
Percentage Mediated	2.2648	0.3033	1.6704	2.8591	7.47	<.0001
Percentage Due to Interaction	2.3889	0.2256	1.9467	2.8311	10.59	<.0001
Percentage Eliminated	3.8088	0.4133	2.9987	4.6189	9.22	<.0001

OUTCOME MODEL
Y = BPBI

Outcome Model Estimates							
Parameter		Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		-6.7311	0.04221	-6.8138	-6.6484	25430.5282	<.0001
mage_imp3	2	0.2064	0.05256	0.1034	0.3094	15.4200	<.0001
mage_imp3	1	0.1445	0.04401	0.05821	0.2307	10.7750	0.0010
mage_imp3	0	0	.	.	.	.	.
mhisp	Non-Hispanic	-0.1657	0.02391	-0.2125	-0.1188	48.0150	<.0001
mhisp	Hispanic	0	.	.	.	.	.
parity2	1	-0.08358	0.02677	-0.1361	-0.03111	9.7470	0.0018
parity2	2	-0.2685	0.03469	-0.3365	-0.2006	59.9308	<.0001
parity2	0	0	.	.	.	.	.
black	1 OR = 1.595	0.4669	0.04544	0.3778	0.5560	105.5560	<.0001
black	0	0	.	.	.	.	.
dm	1 OR = 4.28	1.4544	0.07355	1.3103	1.5986	391.0716	<.0001
dm	0	0	.	.	.	.	.

MEDIATOR MODEL
Y=DIABETES

Mediator Model Estimates							
Parameter		Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		-6.6138	0.04051	-6.6932	-6.5344	26658.4873	<.0001
mage_imp3	2	2.3456	0.04199	2.2633	2.4279	3121.1015	<.0001
mage_imp3	1	1.4625	0.04106	1.3820	1.5429	1268.9032	<.0001
mage_imp3	0	0	.	.	.	.	.
mhisp	Non-Hispanic	-0.4925	0.01101	-0.5141	-0.4709	2001.2500	<.0001
mhisp	Hispanic	0	.	.	.	.	.
parity2	1	0.1491	0.01254	0.1245	0.1737	141.3720	<.0001
parity2	2	0.4252	0.01332	0.3991	0.4513	1019.0191	<.0001
parity2	0	0	.	.	.	.	.
black	1 OR = 1.55	0.4391	0.02237	0.3953	0.4830	385.2443	<.0001
black	0	0	.	.	.	.	.

MEDIATOR: SHOULDER DYSTOCIA

Summary of Effects						
	Estimate	Standard Error	Wald 95% Confidence Limits		Z	Pr > |Z|
Odds Ratio Total Effect	1.6137	0.07416	1.4683	1.7590	8.27	<.0001
Odds Ratio Controlled Direct Effect (CDE)	1.6496	0.07539	1.5019	1.7974	8.62	<.0001
Odds Ratio Natural Direct Effect (NDE)	1.6496	0.07539	1.5019	1.7974	8.62	<.0001
Odds Ratio Natural Indirect Effect (NIE)	0.9782	0.004886	0.9686	0.9878	-4.46	<.0001
Total Excess Relative Risk	0.6137	0.07416	0.4683	0.7590	8.27	<.0001
Excess Relative Risk Due to CDE	0.5527	0.06415	0.4270	0.6784	8.62	<.0001
Excess Relative Risk Due to NDE	0.6496	0.07539	0.5019	0.7974	8.62	<.0001
Excess Relative Risk Due to NIE	-0.03594	0.008229	-0.05207	-0.01981	-4.37	<.0001
Percentage Mediated	-5.8560	1.4560	-8.7097	-3.0024	-4.02	<.0001
Percentage Due to Interaction	13.4887	0.5314	12.4473	14.5302	25.39	<.0001
Percentage Eliminated	9.9388	1.2496	7.4896	12.3880	7.95	<.0001

OUTCOME MODEL

Outcome Model Estimates							
Parameter		Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		-6.8417	0.04258	-6.9251	-6.7582	25818.7053	<.0001
mage_imp3	2	0.2340	0.05261	0.1309	0.3371	19.7814	<.0001
mage_imp3	1	0.1381	0.04415	0.05155	0.2246	9.7821	0.0018
mage_imp3	0	0	.	.	.	.	.
mhisp	Non-Hispanic	-0.2146	0.02402	-0.2617	-0.1675	79.7811	<.0001
mhisp	Hispanic	0	.	.	.	.	.
parity2	1	-0.09460	0.02691	-0.1473	-0.04186	12.3600	0.0004
parity2	2	-0.2222	0.03481	-0.2904	-0.1540	40.7523	<.0001
parity2	0	0	.	.	.	.	.
black	1 OR = 1.65	0.5005	0.04570	0.4110	0.5901	119.9695	<.0001
black	0	0	.	.	.	.	.
shoulder_dys	1 OR = 71.52	4.2706	0.03364	4.2047	4.3365	16115.9617	<.0001
shoulder_dys	0	0	.	.	.	.	.

MEDIATOR MODEL: Y = SHOULDER DYSTOCIA

Mediator Model Estimates							
Parameter		**Estimate**	**Standard Error**	**Wald 95% Confidence Limits**		**Wald Chi-Square**	**Pr > ChiSq**
Intercept		-6.2432	0.03118	-6.3043	-6.1821	40086.2842	<.0001
mage_imp3	**2**	0.04059	0.03834	-0.03455	0.1157	1.1212	0.2897
mage_imp3	**1**	0.1295	0.03213	0.06657	0.1925	16.2586	<.0001
mage_imp3	**0**	0	.	.	.	.	.
mhisp	**Non-Hispanic**	0.2842	0.01707	0.2507	0.3176	277.0998	<.0001
mhisp	**Hispanic**	0	.	.	.	.	.
parity2	**1**	0.09125	0.01804	0.05589	0.1266	25.5800	<.0001
parity2	**2**	-0.2791	0.02535	-0.3287	-0.2294	121.1597	<.0001
parity2	**0**	0	.	.	.	.	.
black	**1 OR = 0.85**	-0.1582	0.03783	-0.2323	-0.08404	17.4829	<.0001
black	**0**	0	.	.	.	.	.

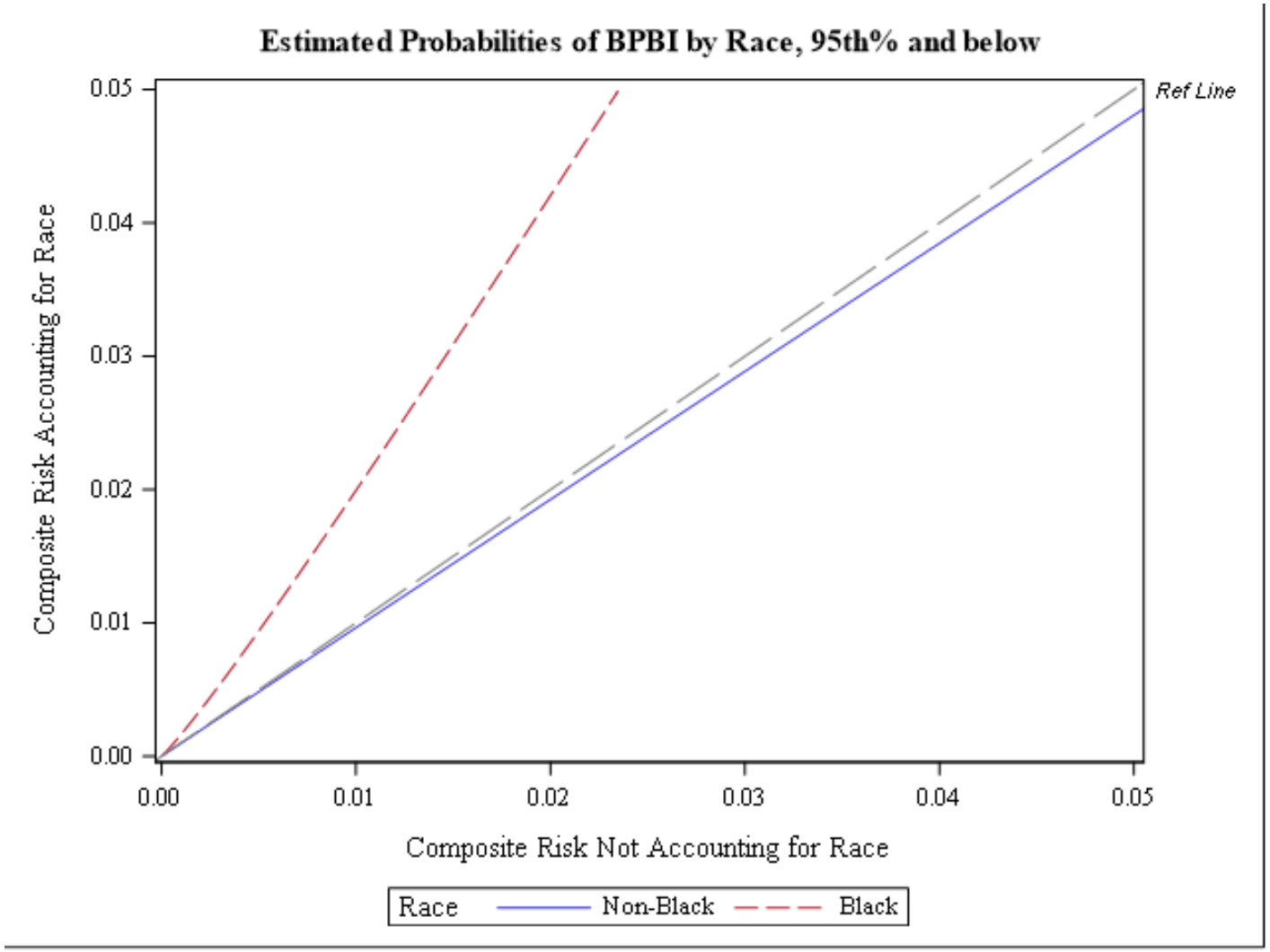
USING PREDICTED PROBABILITIES AS MEDIATORS

```
title "Getting Etas for Black";
proc logistic data=live.live95;
class obese (ref='0') dm (ref='0') dmgest (ref='0') iapc (ref='0')
shoulder_dys (ref='0') macro (ref='0') vagdel (ref='0')
mom_ethnicity (ref='0') mage_imp3 (ref='0’) parity2 (ref='2')
prolong_2nd (ref='0')/param=ref;
model live_bpi (event='1') = obese dm dmgest IAPC shoulder_dys macro
vagdel mom_ethnicity mage_imp3 parity2 prolong_2nd;
output out=black_phats predicted=phat xbeta=eta;
run;
```

INTERACTION BETWEEN RACE AND PHAT

```
title "Getting Eta by Race Interaction";
proc logistic data=black_phats;
class black (ref='0') /param=ref;
model live_bpi (event='1') = black | phat;
run;
```

THE ETAS BY RACE



USING PREDICTED PROBABILITIES AS MEDIATORS

```
title "Mother's Black Race and Eta as Mediator";
PROC CAUSALMED DATA=black_phats pALL ALPHA=.05;
CLASS live_bpi (descending) black (descending);
MODEL live_bpi = black phat;
MEDIATOR phat  = black;
run;
quit;
```

SAS OUTPUT FOR PHATS

Summary of Effects						
	Estimate	Standard Error	Wald 95% Confidence Limits		Z	Pr > |Z|
Odds Ratio Total Effect	1.5151	0.06724	1.3833	1.6469	7.66	<.0001
Odds Ratio Controlled Direct Effect (CDE)	1.8472	0.08201	1.6865	2.0079	10.33	<.0001
Odds Ratio Natural Direct Effect (NDE)	1.8472	0.08201	1.6865	2.0079	10.33	<.0001
Odds Ratio Natural Indirect Effect (NIE)	0.8202	0.001925	0.8164	0.8240	-93.40	<.0001
Total Excess Relative Risk	0.5151	0.06724	0.3833	0.6469	7.66	<.0001
Excess Relative Risk Due to CDE	0.4690	0.04533	0.3801	0.5578	10.35	<.0001
Excess Relative Risk Due to NDE	0.8472	0.08201	0.6865	1.0079	10.33	<.0001
Excess Relative Risk Due to NIE	-0.3321	0.01528	-0.3620	-0.3021	-21.74	<.0001
Percentage Mediated	-64.4686	5.6349	-75.5127	-53.4244	-11.44	<.0001
Percentage Due to Interaction	43.8613	1.6043	40.7169	47.0056	27.34	<.0001
Percentage Eliminated	8.9605	3.1586	2.7697	15.1512	2.84	0.0046

OUTCOME MODEL

Outcome Model Estimates							
Parameter		**Estimate**	**Standard Error**	**Wald 95% Confidence Limits**		**Wald Chi-Square**	**Pr > ChiSq**
Intercept		-0.01228	0.04051	-0.09169	0.06712	0.0919	0.7618
black	**1 OR = 1.85**	0.6137	0.04439	0.5267	0.7007	191.0740	<.0001
black	**0**	0	.	.	.	.	.
phat	**OR = 2.73**	1.0037	0.006693	0.9906	1.0169	22487.8256	<.0001

MEDIATOR MODEL

Mediator Model Estimates							
Parameter		**Estimate**	**Standard Error**	**Wald 95% Confidence Limits**		**Wald Chi-Square**	**Pr > ChiSq**
Intercept		-7.3405	0.000440	-7.3414	-7.3397	278391561	<.0001
black	**1**	-0.1974	0.001932	-0.2012	-0.1937	10442.9920	<.0001
black	**0**	0	.	.	.	.	.
Scale		1.0741	0.000303	1.0735	1.0747		

CONCLUSION

- We still don't know why Blacks have increased risk of BPBI or what mediates this risk.
- We do know that the pathway posited by conventional wisdom does not appear to be the mechanism for black mothers.
- Next steps: looking at hospital and community level covariates that allow us to tease apart individual risk factors that may be associated with race from hospital and community levels factors associated with race.